CAPSTONE PROJECT LAB · GROUP 9 · FALL 2026

Off-Grid Vehicle Tracker

A fully off-grid system that shows a driver the **direction and distance** to a partner vehicle on a sunlight-readable dashboard display — no cell service, no WiFi, no infrastructure. Designed for convoy travel in remote areas such as Alaska.

LoRa radios running the open-source **Meshtastic** mesh firmware exchange GPS positions between vehicles; a custom-firmware display unit computes bearing and distance and renders a live navigation arrow.

Member	Role
Deo	Protocol firmware — Meshtastic BLE client, protobuf decoding
Josiah	Display & navigation firmware — rendering, bearing/distance math
Victor	Radio & hardware lead — Meshtastic config, antennas, range testing
Cannon	Systems & test lead — vehicle integration, drive tests, reports

Sponsor: Dr. Richard Gale · August 24 – December 10, 2026 · github.com/deosgracius/offgrid-vehicle-tracker

1 · The Problem

Two vehicles traveling together in remote terrain routinely lose sight of each other — and with no cell coverage, there is no way to answer the basic question *"where is my partner?"* Existing options are a poor fit: satellite messengers are subscription-based and not glanceable while driving; phone apps need cell service; amateur-radio APRS needs licensing and equipment. The goal is a self-contained pair of devices: power them from the vehicle, and each driver sees an always-updating arrow and distance to the other vehicle.

2 · System Architecture

Each vehicle carries two devices. A **Heltec WiFi LoRa 32 V4** (with GPS module) runs **stock, unmodified Meshtastic firmware** and broadcasts its position over 915 MHz LoRa. A **Waveshare ESP32-S3 display unit** with a 4.2″ reflective LCD connects to its local Heltec over Bluetooth Low Energy and renders the navigation display. The only custom firmware in the whole system runs on the display unit.

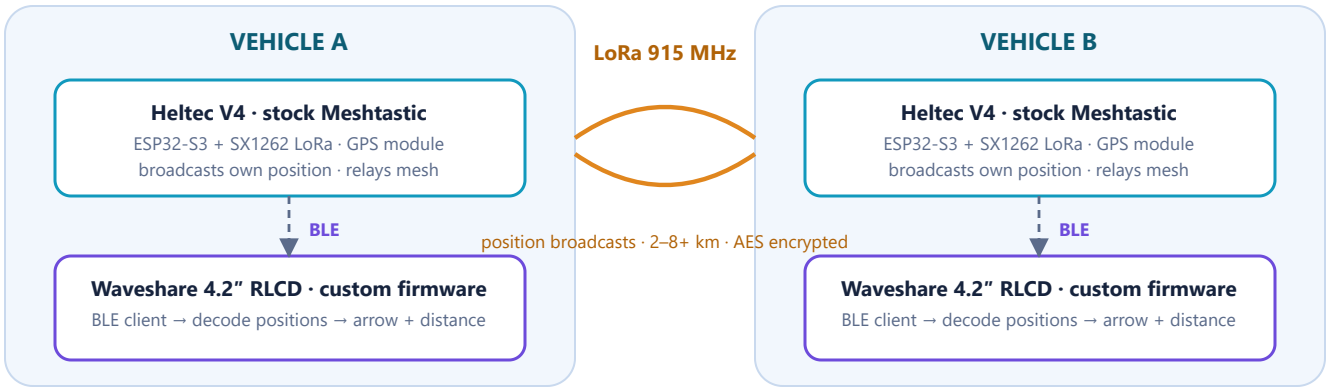


Fig. 1 — System architecture. Identical hardware in each vehicle; positions flow both directions over LoRa.

Key design decision — no Meshtastic "hacking" required. Meshtastic exposes a documented **Client API**: a stream of protobuf messages over BLE, serial, or TCP — the same interface its phone apps use. The device firmware already handles GPS reading, scheduled position broadcasts, AES encryption, mesh relaying, and storing each node's last known position. Our display unit simply connects as a BLE client and consumes `Position` packets. This turns a risky firmware-fork project into a well-scoped client application, and any third Meshtastic node in the area (e.g. one left on a ridge) automatically extends range by relaying.

3 · How It Works — Data Flow

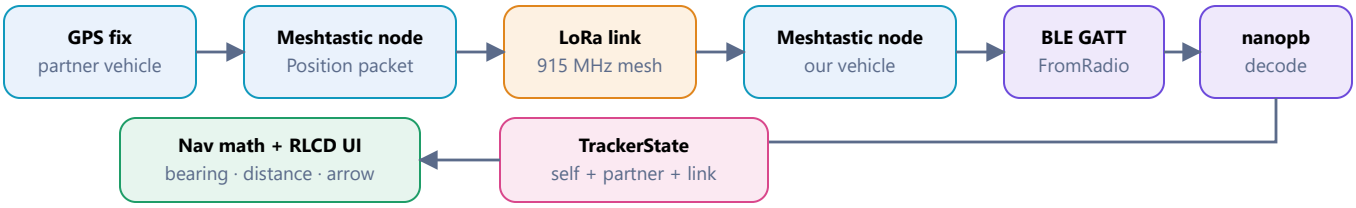


Fig. 2 — Data path from the partner's GPS fix to the arrow on our display. Our own position takes the same path without the LoRa hop.

The boundary between the two firmware halves is a single struct — the **contract** between the protocol module (Deo) and the UI module (Josiah), letting both develop in parallel from week 2:

```
struct TrackerState {
  Fix  self;      // lat, lon, course-over-ground, speed, fix_ok   (own Heltec's GPS)
  Fix  partner;   // lat, lon, age_seconds, valid                  (from mesh Position packets)
  bool ble_connected;
};
```

4 · Hardware

Component	Key specs	Qty
Heltec WiFi LoRa 32 V4 radio + GPS node	ESP32-S3 · SX1262 LoRa, up to 28 dBm TX / −137 dBm sensitivity · WiFi + BLE · 0.96″ OLED · USB-C · solar-charge input · GNSS module on 8-pin connector · Meshtastic-supported	2 (1 per vehicle)
Waveshare ESP32-S3-RLCD-4.2 driver display	4.2″ reflective LCD, 300×400 B/W — sunlight-readable, e-paper look with fast refresh · ESP32-S3, 16 MB flash / 8 MB PSRAM · BLE + WiFi · USB-C, 18650 option, microSD, RTC	1–2
External 915 MHz antennas + roof mounts	The single biggest range factor — stock whip antennas inside a metal vehicle severely reduce range	2
12 V→USB power	Vehicle power for both units; no battery management needed while driving	2–4

Week-1 hardware checks. (1) The V4 board has a GNSS *connector*, not onboard GPS — confirm the purchased kits include the GNSS module. (2) The reflective LCD is excellent in daylight but has no backlight — night driving needs a small edge light. (3) The V4 is a new (late-2025) board — confirm the current Meshtastic release fully supports it with the GNSS module.

5 · Navigation: Direction & Distance

Distance comes from the haversine formula between our fix and the partner's last fix. **Bearing to partner** is the standard initial-bearing formula, referenced to true north:

$$\theta = \operatorname{atan2}\big(\sin\Delta\lambda \cdot \cos\phi_2,\ \cos\phi_1 \cdot \sin\phi_2 - \sin\phi_1 \cdot \cos\phi_2 \cdot \cos\Delta\lambda\big)$$

But the driver doesn't think in compass degrees — the arrow must point **relative to the direction the vehicle is facing**. We get vehicle heading from GPS **course-over-ground**, which is accurate above ~5 mph — no magnetometer needed (compasses are unreliable inside vehicles anyway). The arrow angle is simply:

```
arrow = bearing_to_partner − own_course
```

When stationary or very slow, the display falls back to an absolute compass rose (N/E/S/W) or holds the last good heading.

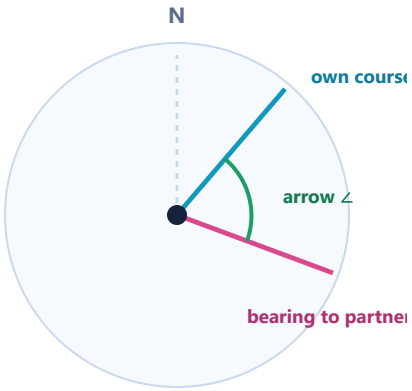


Fig. 3 — The displayed arrow angle is the bearing minus the vehicle's own course.

The driver's display

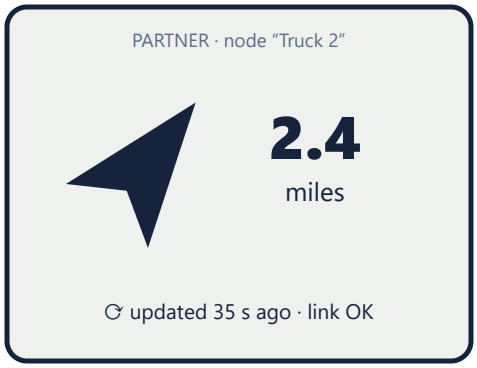


Fig. 4 — Display concept: glanceable arrow, distance, and data age.

Data age is a safety feature: with 30–60 s broadcast intervals and possible dropouts, the driver must always know how fresh the fix is.

Range expectations

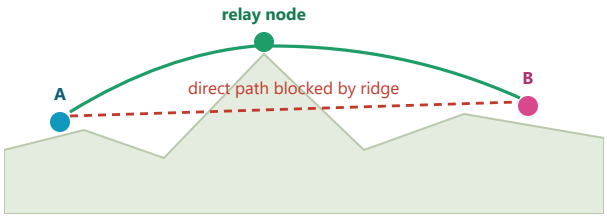
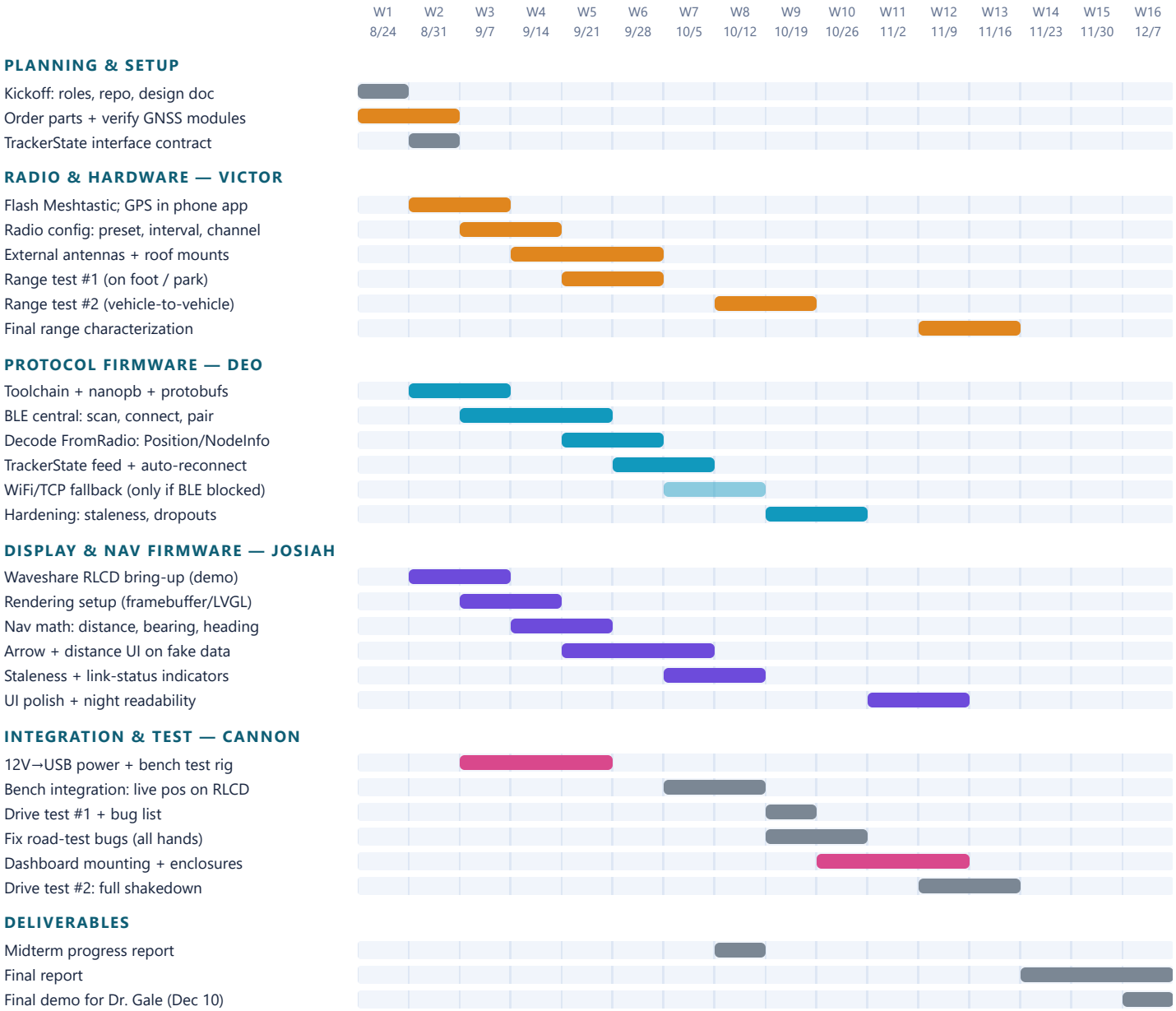


Fig. 5 — Terrain dominates range. A third Meshtastic node on high ground relays automatically.

Vehicle-to-vehicle at ground level, realistic range is **2–8 km** in open or rolling terrain — much more with line of sight, less in deep valleys or dense forest. Mitigations: roof-mounted antennas (largest single win), slower/longer-range LoRa presets, and mesh relaying.

6 · Semester Timeline

Deo — Protocol firmware · Josiah — Display & nav · Victor — Radio & hardware · Cannon — Systems & test · Team — shared



W13–14 double as schedule buffer (Thanksgiving week has no scheduled build work). The live, editable version of this chart is `planning/Group9_Gantt.html` in the repo.

7 · Key Milestones

When	Milestone
W2 (Sep 4)	All parts in hand; GNSS modules verified; interface contract signed off
W3–W5	**Make-or-break:** BLE client receives and decodes Position packets (Deo). Fallback decision point: switch to WiFi/TCP transport if BLE pairing is still blocked after 2 weeks.
W7 (mid-Oct)	Bench demo: live partner position rendered on the RLCD
W9	First two-vehicle drive test
W12–13	Road-tested full system; final range numbers
W16 (Dec 10)	Final demo + report for Dr. Gale

8 · Division of Responsibilities

Member	Scope	Primary deliverables
Deo Protocol firmware	Meshtastic BLE client on the display board: NimBLE central, pairing/bonding, protobuf (nanopb) decode of `FromRadio`, node tracking, reconnect logic. Highest-uncertainty track — starts the day boards arrive.	Working `TrackerState` feed; BLE pairing writeup; fallback decision by W5
Josiah Display & nav	RLCD driver + rendering, arrow/distance/staleness UI, haversine + bearing + course-over-ground heading. Develops against simulated data — zero dependency on Deo's track until integration (W7).	Driver UI; nav math module with unit tests against known coordinates
Victor Radio & hardware	Meshtastic flashing/config on both Heltecs, GNSS integration, antenna selection and mounting, structured range testing. Owns the “how far does it really work” answer.	Radio config baseline; antenna recommendation; dated range-test logs
Cannon Systems & test	Vehicle power, dashboard mounting, enclosures, night lighting; integration and drive testing; parts ordering/tracking; report and presentation assembly.	Installed, road-tested units; parts ledger; midterm + final reports

9 · Risks & Mitigations

Risk	Impact	Mitigation
BLE pairing/bonding between our ESP32 and Meshtastic fights back	High	Scheduled first (W3–5); proven prior art exists; hard fallback to Meshtastic's identical API over WiFi/TCP (port 4403) decided at W5
Kits arrive without GNSS modules	High	Verified in week 1; modules ordered separately same week if missing
Parts delay	Medium	Everything ordered W1, including antennas; work that needs no hardware (nav math, UI on fake data) proceeds anyway
Real-world range below expectations	Medium	Roof antennas, longer-range LoRa presets, optional relay node; range facts reported honestly with terrain context
V4 board too new for stable Meshtastic support	Medium	Checked in W2 bench bring-up; V3-pin-compatible fallback hardware exists
RLCD unreadable at night	Low	Small LED edge light in enclosure (Cannon, W10–11)

10 · References

Meshtastic project / Client API docs	meshtastic.org · meshtastic.org/docs/development/device/client-api
Meshtastic protobuf definitions	github.com/meshtastic/protobufs
Heltec WiFi LoRa 32 V4	heltec.org/project/wifi-lora-32-v4
Waveshare ESP32-S3-RLCD-4.2 (docs + demo code)	waveshare.com/esp32-s3-rlcd-4.2.htm · github.com/waveshareteam/ESP32-S3-RLCD-4.2
ESP32-as-BLE-client prior art	github.com/0io/ESPHome-Meshtastic-MQTT
Group 9 repository	github.com/deosgracius/offgrid-vehicle-tracker